

Quantitative Analysis

Course: CPA — Certified Public Accountant

Level: Foundation Level (CA15)

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Syllabus version: Based on the standard KASNEB CPA Foundation Level Quantitative Analysis syllabus topics (CA15) — basic algebra, financial mathematics (simple/compound interest, annuities), descriptive statistics, and elementary probability. Authored from general quantitative-methods knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

Contents

1. Basic algebra and linear equations
 2. Simple and compound interest
 3. Annuities, basics
 4. Measures of central tendency
 5. Measures of dispersion
 6. Index numbers
 7. Basic probability
 8. Simple linear regression and correlation, introduction
 9. Full worked question — annuity and standard deviation
 10. Practice examination — KASNEB-style paper (original, not a past paper)
 11. Marking guide — model answers to the practice examination
 12. Exam technique and revision checklist
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1. Basic algebra and linear equations

Many quantitative business problems reduce to a linear equation that can be solved by isolating the unknown. For example, given $\text{Total cost} = \text{Fixed cost} + (\text{Variable cost per unit} \times \text{Units})$, if total cost, fixed cost and variable cost per unit are known, the number of units can be found by:

$$\text{Units} = (\text{Total cost} - \text{Fixed cost}) \div \text{Variable cost per unit}$$

2. Simple and compound interest

Simple interest: $I = P \times R \times T$ (Principal $\times$ Rate $\times$ Time), calculated only on the original principal each period.

Compound interest: $A = P \times (1 + r)^n$, where interest is calculated on the principal plus previously added interest — "interest on interest" — so compound amounts grow faster than simple interest over time.

3. Annuities, basics

An **annuity** is a series of equal periodic payments. The present value of an ordinary annuity (payments at the end of each period) can be computed as:

$$PV = A \times [1 - (1 + r)^{-n}] \div r$$

where **A** is the periodic payment, **r** is the interest rate per period, and **n** is the number of periods. Annuities are commonly used to model loan repayments, regular savings plans, and lease payments.

4. Measures of central tendency

- **Mean** — sum of values $\div$ number of values; sensitive to outliers.
 - **Median** — the middle value of ordered data (or average of the two middle values for an even count); unaffected by outliers.
 - **Mode** — the most frequently occurring value.
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5. Measures of dispersion

- **Range** — highest value minus lowest value; simple but uses only two data points.
- **Variance** — the average of the squared deviations of each value from the mean.

- **Standard deviation** — the square root of the variance; measures how spread out values are around the mean, in the same units as the original data.
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6. Index numbers

A **price index** measures the relative change in price over time compared to a fixed base period (set to 100):

$$\text{Price index} = (\text{Price in current period} \div \text{Price in base period}) \times 100$$

An index of 112 means the value has risen 12% relative to the base period.

7. Basic probability

Probability of an event = Number of favourable outcomes $\div$ Total number of possible outcomes, ranging from 0 (impossible) to 1 (certain). For mutually exclusive events, the probability that either occurs is the sum of their individual probabilities. For independent events, the probability that both occur is the product of their individual probabilities.

8. Simple linear regression and correlation, introduction

Correlation measures the strength and direction of a linear relationship between two variables, typically expressed as a **correlation coefficient (r)** ranging from -1 (perfect negative relationship) through 0 (no linear relationship) to $+1$ (perfect positive relationship). **Simple linear regression** fits a straight line ($y = a + bx$) to describe how a dependent variable (y) tends to change as an independent variable (x) changes, used for description and, cautiously, for forecasting within the range of the observed data.

9. Full worked question — annuity and standard deviation

Scenario. (a) Compute the present value of an ordinary annuity paying KSh 50,000 per year for 3 years, at a discount rate of 10% per year. (b) A data set of five values has a mean of 20. The squared deviations from the mean are: 4, 9, 1, 16, 0. Compute the variance and standard deviation.

Solution:

(a) $PV = 50,000 \times [1 - (1.10)^{-3}] \div 0.10 = 50,000 \times [1 - 0.751315] \div 0.10 = 50,000 \times 2.48685 = \text{KSh } 124,343$ (approximately).

(b) Variance = Sum of squared deviations $\div$ Number of values = $(4+9+1+16+0) \div 5 = 30 \div 5 = 6$.

Standard deviation = $\sqrt{6} = \approx 2.449$.

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB CPA examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

CPA FOUNDATION LEVEL — QUANTITATIVE ANALYSIS (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

A trader's total cost is given by: Total cost = 60,000 + (300 $\times$ units). Last month, total cost was KSh 165,000.

- (a) Form the equation and solve for the number of units. (8 marks)
 - (b) Compute the compound amount of KSh 80,000 invested at 8% per year for 3 years. (8 marks)
 - (c) Briefly distinguish simple interest from compound interest. (4 marks)
- (Total: 20 marks)
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QUESTION TWO

- (a) Compute the present value of an ordinary annuity paying KSh 40,000 per year for 4 years, at a discount rate of 12% per year. (12 marks)
 - (b) Outline TWO practical uses of annuity calculations in business/personal finance. (8 marks)
- (Total: 20 marks)
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QUESTION THREE

The weekly sales (KSh '000) of a shop over six weeks are: 40, 35, 40, 50, 45, 40.

(a) Compute the mean. (5 marks)

(b) Compute the median. (5 marks)

(c) Identify the mode. (4 marks)

(d) Compute the range. (6 marks)

(Total: 20 marks)

QUESTION FOUR

(a) A commodity's price was KSh 60 in the base year and KSh 75 in the current year. Compute and interpret the price index. (8 marks)

(b) A box contains 5 white balls and 7 black balls. One ball is drawn at random. Compute the probability it is (i) white, (ii) black. (8 marks)

(c) Two independent events, A and B, have probabilities 0.4 and 0.5 respectively. Compute the probability that both occur. (4 marks)

(Total: 20 marks)

QUESTION FIVE

(a) Explain what a correlation coefficient measures, and interpret a value of -0.9 . (10 marks)

(b) Explain the purpose of simple linear regression, and one caution when using it for forecasting. (10 marks)

(Total: 20 marks)

11. Marking guide — model answers to the practice examination

QUESTION ONE

(a) (8 marks) — $165,000 = 60,000 + 300x \rightarrow 300x = 105,000 \rightarrow x = \mathbf{350 \text{ units}}$.

(b) (8 marks) — $A = 80,000 \times (1.08)^3 = 80,000 \times 1.259712 = \mathbf{KSh 100,777}$ (approximately).

(c) (4 marks) — Simple interest is calculated only on the original principal each period; compound interest is calculated on the principal plus previously added interest, growing faster over time.

QUESTION TWO

(a) (12 marks) — $PV = 40,000 \times [1 - (1.12)^{-4}] \div 0.12 = 40,000 \times [1 - 0.635518] \div 0.12 = 40,000 \times 3.03735 = \text{KSh } 121,494$ (approximately).

(b) (8 marks — 4 marks each) — Modelling loan repayments; valuing regular savings plans or lease payments (any two).

QUESTION THREE

(a) (5 marks) — Mean = $(40+35+40+50+45+40) \div 6 = 250 \div 6 = \text{KSh } 41,667$ (approximately).

(b) (5 marks) — Ordered: 35,40,40,40,45,50 → Median = $(40+40) \div 2 = \text{KSh } 40,000$.

(c) (4 marks) — Mode = **KSh 40,000** (occurs three times).

(d) (6 marks) — Range = $50 - 35 = \text{KSh } 15,000$.

QUESTION FOUR

(a) (8 marks) — Price index = $(75 \div 60) \times 100 = 125$; the price has risen by 25% relative to the base year.

(b) (8 marks) — Total balls = 12. (i) $P(\text{white}) = 5/12 = \approx 0.417$. (ii) $P(\text{black}) = 7/12 = \approx 0.583$.

(c) (4 marks) — $P(A \text{ and } B) = 0.4 \times 0.5 = 0.20$.

QUESTION FIVE

(a) (10 marks) — A correlation coefficient measures the strength and direction of a linear relationship between two variables, ranging from -1 to $+1$; a value of -0.9 indicates a strong negative (inverse) linear relationship — as one variable increases, the other tends to decrease strongly.

(b) (10 marks) — Simple linear regression fits a straight line describing how a dependent variable changes as an independent variable changes, useful for description and cautious forecasting; a key caution is that forecasting outside the range of the original observed data (extrapolation) can be unreliable, since the relationship may not hold beyond the data actually observed.

12. Exam technique and revision checklist

- Word-problem/equation questions: always write the equation explicitly before solving.
- Interest questions: state clearly whether simple or compound, and use the matching formula.
- Annuity questions: state the formula explicitly and be careful with the negative exponent in $(1+r)^{-n}$.

- Central tendency/dispersion questions: order the data first, and double check for ties in the mode.
- Correlation/regression questions: always interpret the number in plain words (strength AND direction) when asked to interpret, not just state the formula.

Quick revision — one line per topic:

1. Simple interest: $I = PRT$. Compound interest: $A = P(1+r)^n$.
2. Annuity $PV = A \times [1 - (1+r)^{-n}] \div r$.
3. Mean = $\text{sum} \div \text{count}$; median = middle value (ordered); mode = most frequent value.
4. Range = highest – lowest; variance = average squared deviation from mean; standard deviation = $\sqrt{\text{variance}}$.
5. Price index = $(\text{Current price} \div \text{Base price}) \times 100$.
6. Probability = favourable outcomes $\div$ total outcomes; independent events multiply, mutually exclusive events add.
7. Correlation coefficient: -1 to $+1$, measures strength and direction of a linear relationship.
8. Linear regression: $y = a + bx$; caution against extrapolating beyond the observed data range.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.